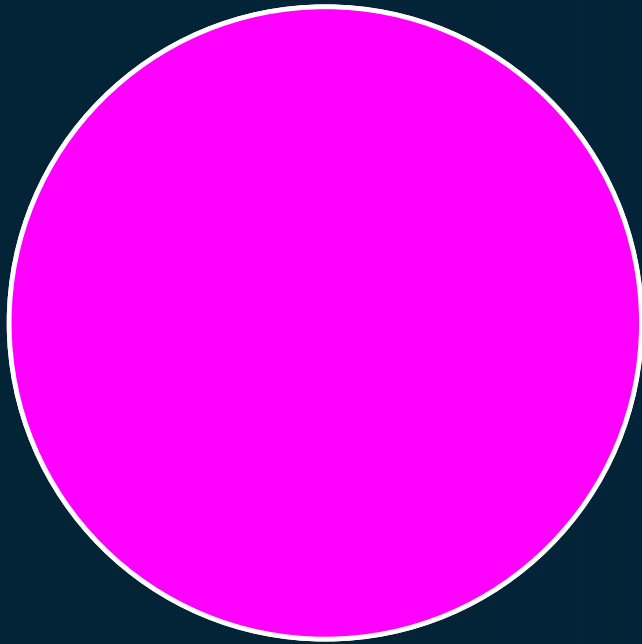


AI Data Processing & Integrity Controls

Chapter 04 · Episode 01 · Domain 1.2 — Data Integrity Is Model Integrity



Learning Objectives

1. Explain all seven data-processing controls and their function
2. Connect data-quality failures to model security outcomes
3. Apply data integrity controls to a training-pipeline scenario

Why Data Security IS Model Security

TRADITIONAL APP SECURITY

Vulnerability lives in code

Fix the code → patch

Auditable, point-in-time

Surface: running app

AI DATA-ORIGIN

Vulnerability in the weights

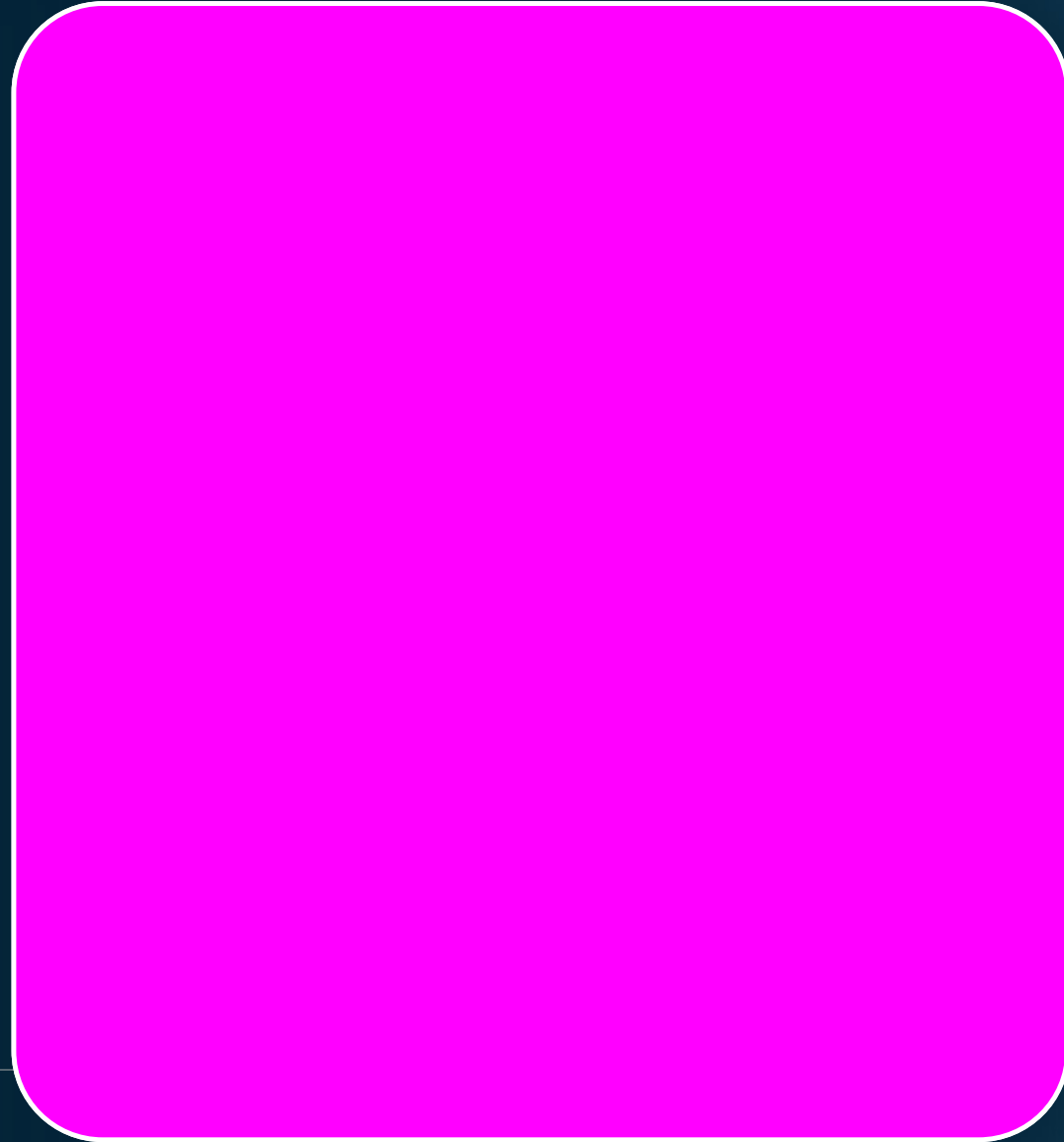
Fix requires full retraining

Opaque; found in production

Surface: the data pipeline

Seven Data-Processing Security Controls

- 1 → Cleansing
- 2 → Verification
- 3 → Lineage
- 4 → Integrity
- 5 → Provenance
- 6 → Augmentation
- 7 → Balancing



Controls 1–4: Foundation of Trust



CLEANSING

Remove injected/malformed
(valid $\neq$ safe)



LINEAGE

Source-to-training audit trail



VERIFICATION

Confirm records are authentic



INTEGRITY

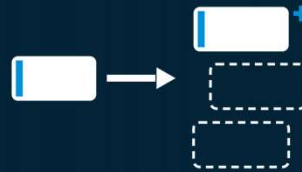
Hashes detect modification

Controls 5–7: Source, Expand, Balance



PROVENANCE

Source legitimacy & consent
EU AI Act / GDPR requirement



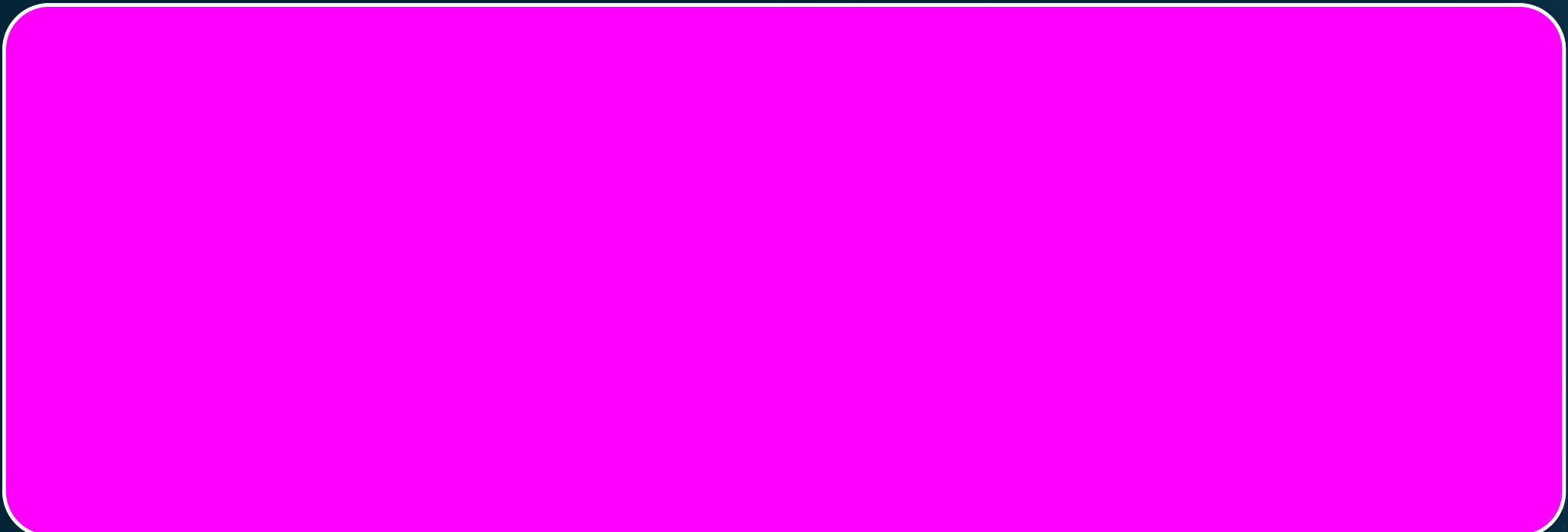
AUGMENTATION

Expand data artificially
Review for distortion & bias



BALANCING

Proportional classes
Imbalance = blind spots



Demo: Data Integrity Pipeline Checker

Seven checks in sequence — an integrity-hash failure blocks the pipeline (trust score insufficient)

Control → Security Outcome



CLEANSING

Prevents garbage models



INTEGRITY

Detects tampering



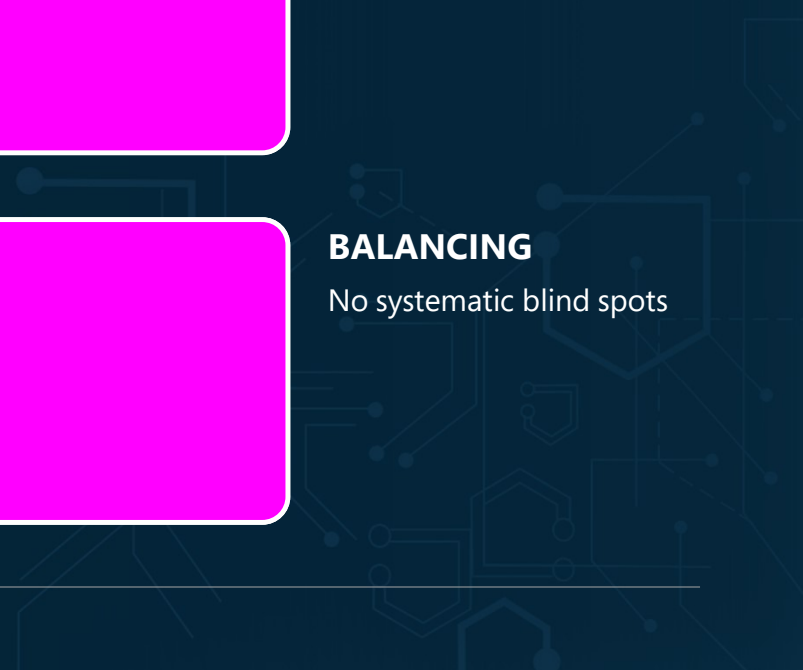
VERIFICATION

No false ground truth



BALANCING

No systematic blind spots



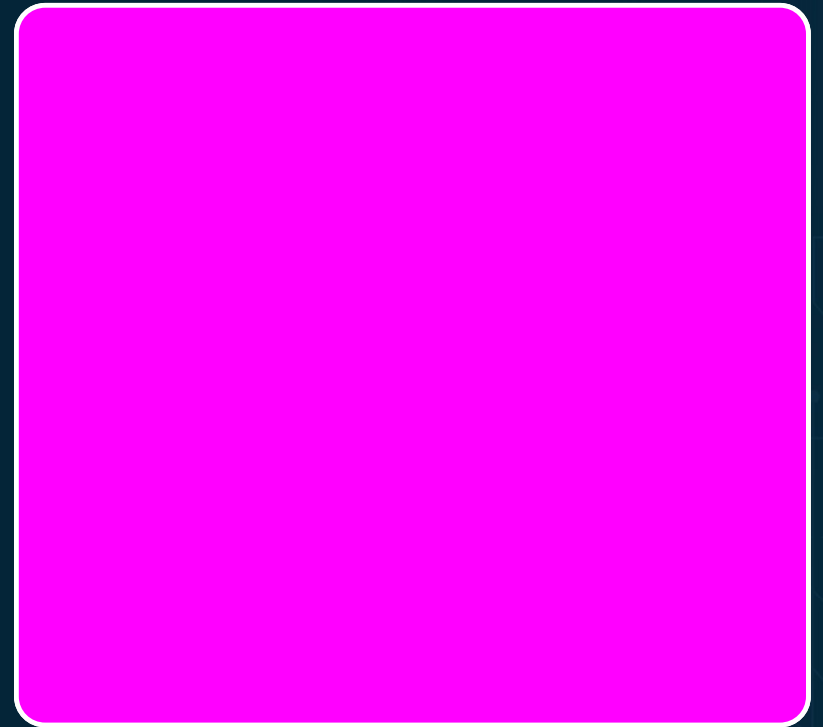
Exam Match Rule

Scenario → which control applies

Missed attack type → balancing / verification

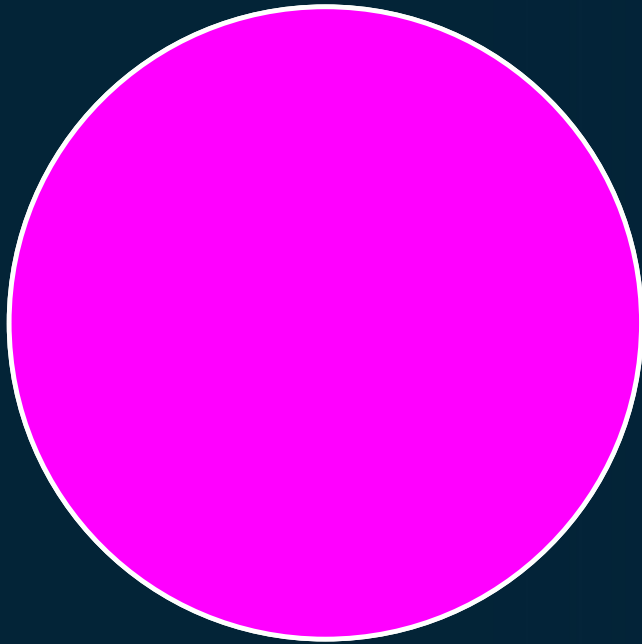
Modified dataset → integrity / lineage

Unauthorized source → provenance



Data Types, RAG, Embeddings & Watermarking

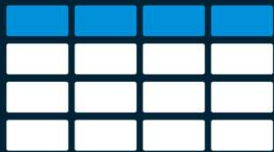
Chapter 04 · Episode 02 · Domain 1.2 — Retrieval & Provenance



Learning Objectives

1. Classify structured, semi-structured & unstructured data + risks
2. Trace a RAG query end-to-end, security concern at each step
3. Explain embeddings, vector stores & watermarking

Three AI Data Types



STRUCTURED

Rows/columns, DBs, logs
PII memorization risk



SEMI

JSON/XML, telemetry
Schema fools cleansing



UNSTRUCTURED

Text/images/audio
Highest — extractable

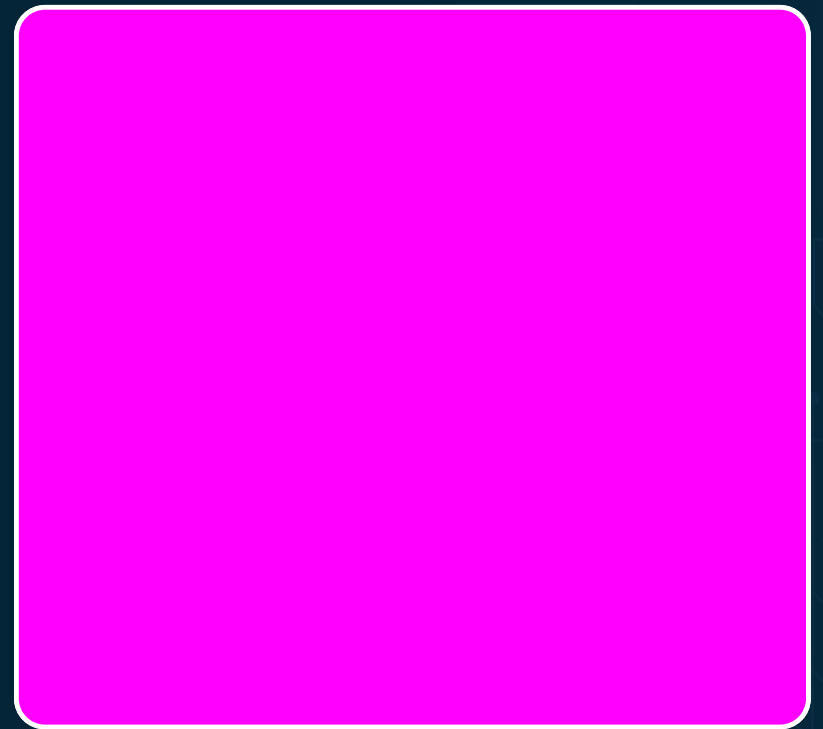
RAG — Retrieval-Augmented Generation

Augments responses with retrieved knowledge at query time

Update the knowledge base without retraining

Keeps sensitive data in a DB, not the weights

Query → embedding → retrieve → augment
→ generate



RAG's Attack Surfaces



QUERY

Crafted query targets sensitive docs



AUGMENTATION

Retrieved content unlabeled/untrusted



RETRIEVAL

RAG poisoning of the knowledge base



GENERATION

Leaks retrieved sensitive content

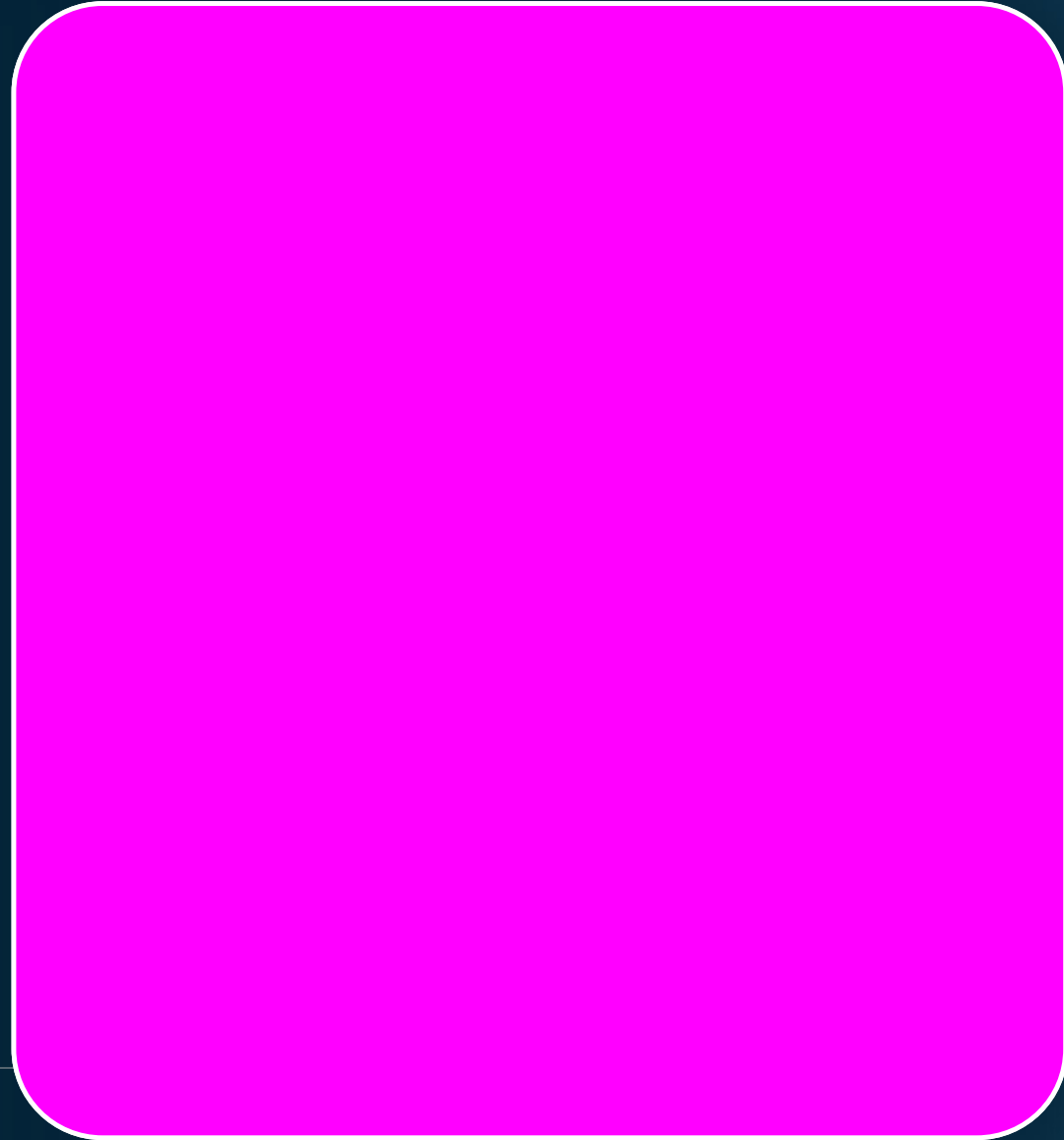
Embeddings & Vector Stores

Vectors encode semantic meaning

'malware' sits near 'ransomware', far from 'sandwich'

Vector stores index them for similarity search

Access control matters as much as the source docs



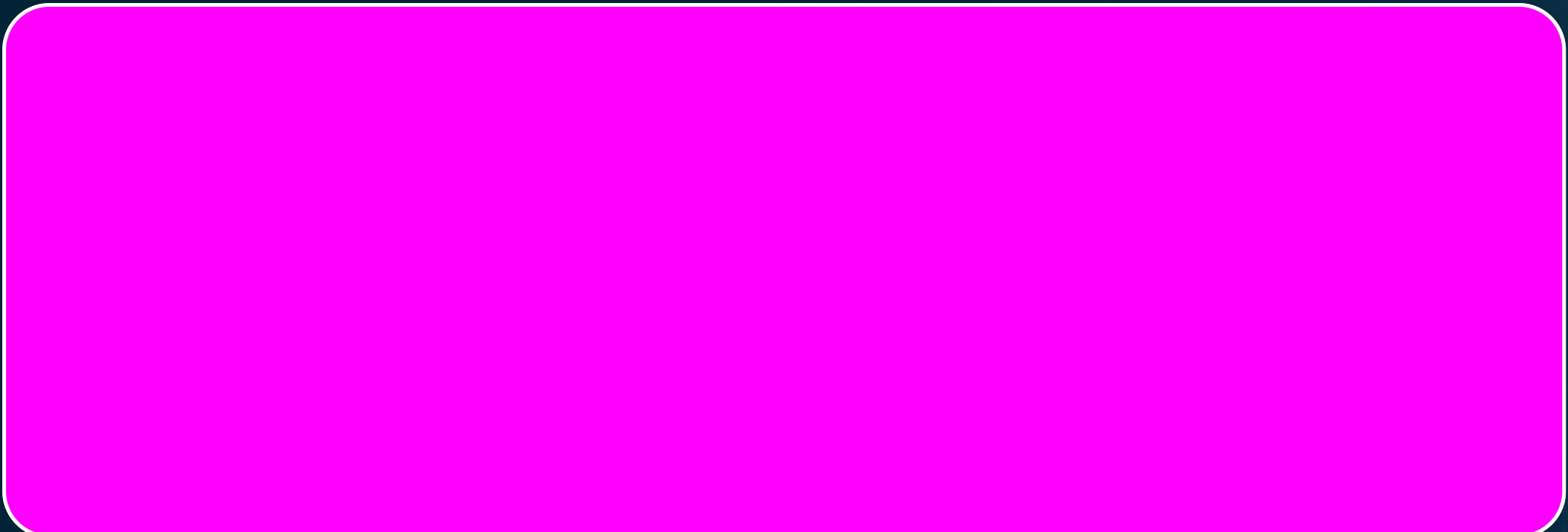
Watermarking — Two Provenance Applications

TRAINING-DATA WATERMARK

Markers embedded pre-training
Model reproduces them in output
Proves it trained on your data
Detects unauthorized dataset use

OUTPUT WATERMARK

Imperceptible signal in output
Text, images, audio
Flags content as AI-generated
Traces it to the model



Demo: RAG Pipeline Security Walkthrough

Query → embedding → retrieval → augmentation → generation, with the control at each step

Security Reference Map

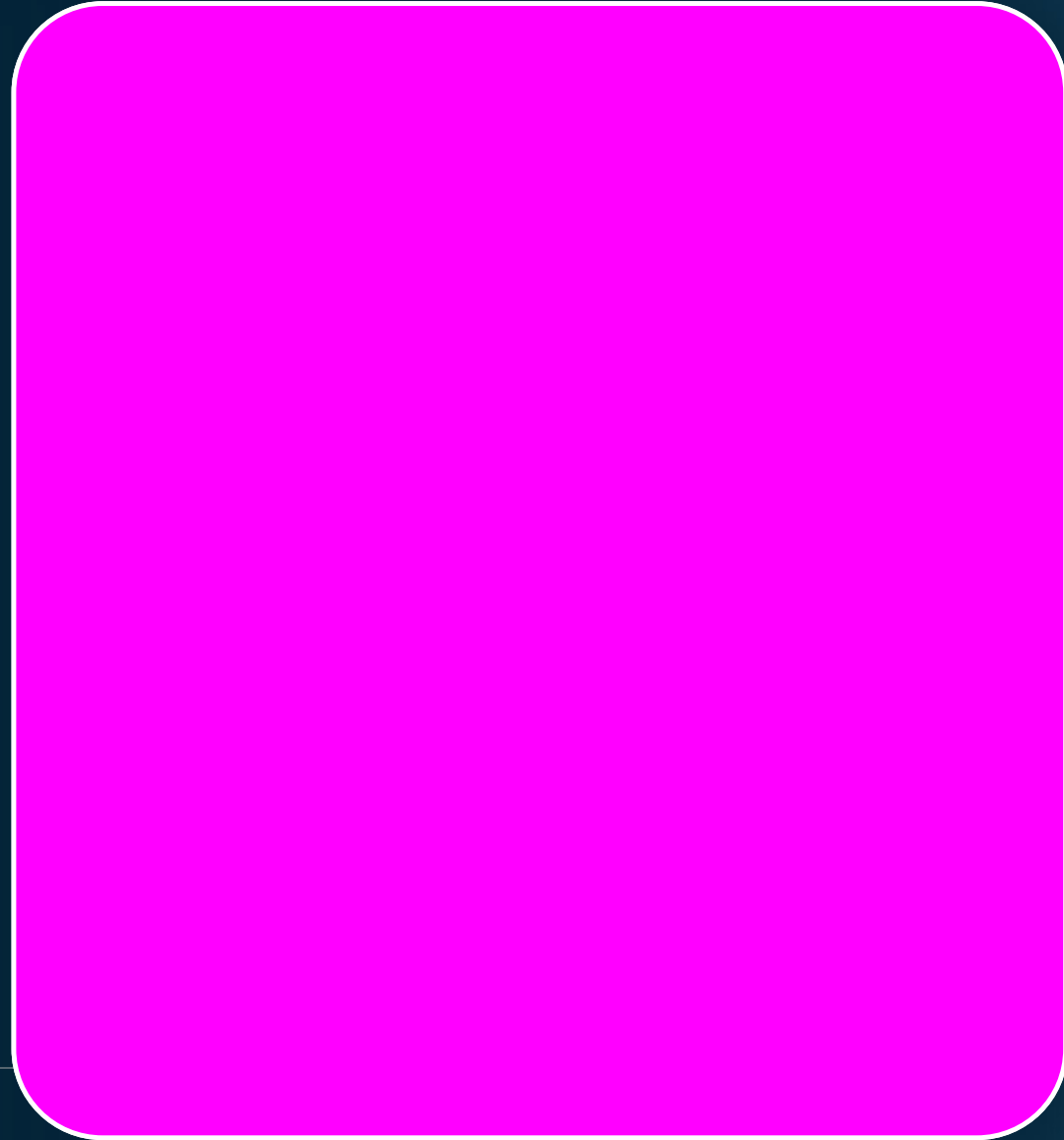
Structured, semi-structured, unstructured —
know the difference

RAG: five steps, an attack surface at each one

Embeddings encode semantic meaning;
vector stores need access control

Watermarking is a provenance control, not
access control

RAG poisoning: knowledge-base access
control and content validation, not prompt
templates alone



Exam Tips



DATA TYPES

Unstructured = highest risk

VECTOR STORES

Need access control

RAG POISONING

Indirect injection via the KB

WATERMARKING

A provenance control